MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**November, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**C/FC/PE/OE**
Course Name	**Design of prestressed concrete structures**	**Course Code**	**CIE 3124**
Semester	**V**	**Max Marks**	**50**
Date	**30/11/2024**	**Duration**	**3 hrs**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	Explain the following: a) Thermoelectric prestressing b) Chemical prestressing	4	1	1,2,8,12	2
1B	A simply supported Type 2 pre-tensioned PSC beam, having a span of 10m, is pre-stressed with a parabolic cable with an eccentricity of 250mm at the center, and a pre-stressing force induced is 900 kN. The cross-section of the beam is an unsymmetrical I-section with a top flange of 500mm x 100 mm, a web of 100mm x 600mm, and a bottom flange of 300mm x 250mm. Take Y_t = 477.7mm and I = 2.08 x 10^{10}mm^4. The working load is 15 kN/m throughout the span. Loss of pre-stress is 20%. **Illustrate and plot the stress distribution diagram at the mid-span section at the working stage and check whether stresses are within the limits as per codal provisions. Use M45 grade of concrete.**	6	2	1,2,3,8,12	3
2A	A PSC beam is designed to carry a live load of 15 kN/m over a span of 15 m. Grade of Concrete: M-40; Characteristic Strength of Steel: 1750 N/mm^2; Area of steel: 800mm^2; Effective pre-stress: 1000 kN; Cable profile: Parabolic which is concentric at	6	3	1,2,12	3

	supports and having an eccentricity of 250mm at mid-span; Effective cover to longitudinal reinforcement: 409.62mm. The beam cross section has top flange: 600mm × 250mm, web: 250mm × 600mm, and bottom flange: 300mm × 300mm, with y_t=490.38mm, I = 4.99×10^{10} mm^4. If the beam is post-tensioned, examine for the **limit state of collapse in flexure as per codal provisions**.				
2B	Demonstrate how failure due to Bonding is controlled in pretensioned members.	**4**	4	1,2,12	**3**
3A	Demonstrate the type of composite construction in prestressed concrete members.	**4**	5	1,2,12	**3**
3B	A PSC beam is designed to carry a live load of 15 kN/m over a span of 15 m. Grade of Concrete: M-40; Characteristic Strength of Steel: 1750 N/mm^2; Area of steel: 800mm^2; Effective pre-stress: 1000 kN; Cable profile: Parabolic which is concentric at supports and having an eccentricity of 250mm at mid-span; Effective cover to longitudinal reinforcement: 409.62mm. The beam cross section has top flange: 600mm × 250mm, web: 250mm × 600mm, and bottom flange: 300mm × 300mm, with y_t=490.38mm, I = 4.99×10^{10} mm^4. If beam is post-tensioned, **design for Shear reinforcement** using 8mm diameter HYSD bars, by examining shear resistance at section uncracked in flexure only.	**6**	3	1,2,12	**3**
4A	A rectangular PSC beam of cross section 350mm wide and 700mm deep has a span of 12m. It is pre-stressed with an initial pre-stressing force of 700kN with the loss of prestress at 20%. The beam carries working load of 8kN/m throughout the span, with creep coefficient =1.5 for M40 grade of concrete. i) Determine the suitable eccentricity for a parabolic cable to limit the deflection under the transfer stage as per codal provisions. ii) Determine the magnitude of uniformly distributed load that can be applied within the serviceability criteria if an eccentricity of 300mm is provided?	**6**	3	1,2,12	**3**
4B	The end block of a pre-stressed concrete beam is rectangular in section 120mm x 300mm. A pre-stressing force of 250kN is transmitted to concrete by a distribution plate l00mm x 150mm, concentrically located at ends of the beam. Apply	**4**	4	1,2,12	**3**

	the codal provisions and d**esign the end block** with Fy-415 grade steel and M-40 grade of concrete.				
5A	Analyse to design a simply supported type-1 post-tensioned PSC beam of span 11m loaded with a working load of 16kN/m. The initial value of pre-stressing force is 900kN, grade of concrete is M40, the characteristic strength of HTS wires used for the tendon is 1600 N/mm^2, and the Loss of pre-stress is 15%. The trial cross-section is as follows: Top flange: 410mm × 150mm, Web: 140mm × 620mm, Bottom flange: 200mm × 250mm, y_t=450.28mm, I = 2.17×10^{10} mm^4.	**8**	2	1,2,3,8,12	**4**
5B	Illustrate the prestressing methods observed during a **field visit to the Netravati bridge construction site** and discuss how they differ. Describe the process of tensioning the steel tendons, including the equipment used with its purpose, and the safety measures implemented during the construction process.	**2**	2	1,2,3,8,12	**3**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***4***
2	***16***
3	***18***
4	***8***
5	***4***

BL	Total Marks
2	**4**
3	**38**
4	**8**
5	